

Titre du projet : Uncertainty-aware anomaly detection & diagnosis in EM-Enabled greenhouses based on probabilistic AI models

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1 DESCRIPTION

With the continuous growth of the global population and the increasing demand for fresh and sustainable products, greenhouse farming is considered one of the key solutions to ensure food security and enhance agricultural productivity [1]. Greenhouses enable year-round crop production, independently of seasonal and climatic conditions, especially in regions where open-field cultivation is limited or risky. However, maintaining suitable indoor conditions requires the continuous operation of heating, ventilation, dehumidification, and lighting systems, which consume substantial energy and represent a significant part of greenhouse operating costs [1]. Anomalies affecting these systems may lead to energy losses, increased production costs, unsuitable growing conditions, equipment malfunction, and reduced product quality. Therefore, accurate anomaly detection has become an important requirement for reliable and energy-efficient greenhouse operation.

Nevertheless, anomaly detection in greenhouses remains challenging because environmental variables and sensor measurements are affected by natural fluctuations, changing weather conditions, noise, and complex physical interactions between equipment and indoor climate variables. Conventional data-driven methods may interpret these normal variations as anomalies, resulting in false alarms and unreliable decisions [2]. Moreover, detecting an anomaly alone is not sufficient, since greenhouse operators also need to identify its root cause and determine whether it originates from an actuator fault, a sensor fault, a control problem, or an abnormal environmental condition. This doctoral project aims to develop an integrated framework for anomaly detection and diagnosis in smart greenhouses by combining physics-informed probabilistic modeling, uncertainty quantification, causal reasoning, and computationally efficient artificial intelligence methods.

2 OBJECTIF PRINCIPAL

Develop and validate an intelligent framework for anomaly detection and diagnosis in smart greenhouses by combining physical system knowledge, uncertainty quantification, causal reasoning, and artificial intelligence methods.

3 SOUS-OBJECTIFS

- Develop a physics-informed probabilistic anomaly detection method that integrates greenhouse physical relationships and quantifies aleatoric and epistemic uncertainty to improve detection reliability and reduce false alarms.
- Develop a causal diagnosis approach to identify the root causes of detected anomalies [3] and distinguish between equipment faults, sensor faults, control problems, and abnormal environmental conditions.
- Develop computationally efficient methods to accelerate anomaly detection and diagnosis for real-time monitoring and decision support in smart greenhouse systems.

4 DESCRIPTION BRÈVE DE LA MÉTHODOLOGIE

The research methodology is organized into four main stages. First, a literature review will be conducted to identify existing anomaly detection and diagnosis methods for greenhouse, agricultural, energy, and cyber-physical systems. The main limitations of current data-driven, physics-based, probabilistic, and causal methods will be analyzed. Second, the normal behavior of the greenhouse will be represented using physical and energetic relationships between external weather conditions, control commands, greenhouse equipment, indoor climate variables, and energy consumption. These relationships will be combined with probabilistic learning methods to detect deviations from normal operation while accounting for aleatoric and epistemic uncertainty. The resulting confidence information will be used to distinguish actual anomalies from natural operating variations and model errors. Third, causal and energetic representations of the greenhouse system will be developed to describe the propagation of effects between weather conditions, actuators, internal greenhouse states, sensors, and energy consumption. After detecting an anomaly, causal inference and root-cause analysis methods will be used to identify its most probable source and distinguish equipment faults from sensor, controller, and environmental anomalies. Finally, computationally efficient artificial intelligence methods will be investigated to accelerate the detection and diagnosis process. The developed framework will be evaluated using data generated by a physics-based greenhouse simulator and, when available, measurements from a real greenhouse test bench. Its performance will be compared with existing data-driven and deterministic methods using detection accuracy, precision, recall, false alarm rate, diagnosis accuracy, uncertainty calibration, and computation time.

5 RÉFÉRENCES (3 ARTICLES DE REVUE)

[1] S. Nagarsheth, K. Agbossou, N. Henao, and M. Bendouma, *The Advancements in Agricultural Greenhouse Technologies: An Energy Management Perspective*, **Sustainability**, vol. 17, no. 8, Art. 3407, 2025.

[2] D. Stjelja, V. Kuzmanovski, R. Kosonen, and J. Jokisalo, *Building Consumption Anomaly Detection: A Comparative Study of Two Probabilistic Approaches*, **Energy and Buildings**, vol. 313, Art. 114249, 2024.

[3] W. Tang, J. Liu, Y. Zhou, and Z. Ding, *Causality-Guided Counterfactual Debiasing for Anomaly Detection of Cyber-Physical Systems*, **IEEE Transactions on Industrial Informatics**, vol. 20, no. 3, pp. 4582–4593, 2024.